

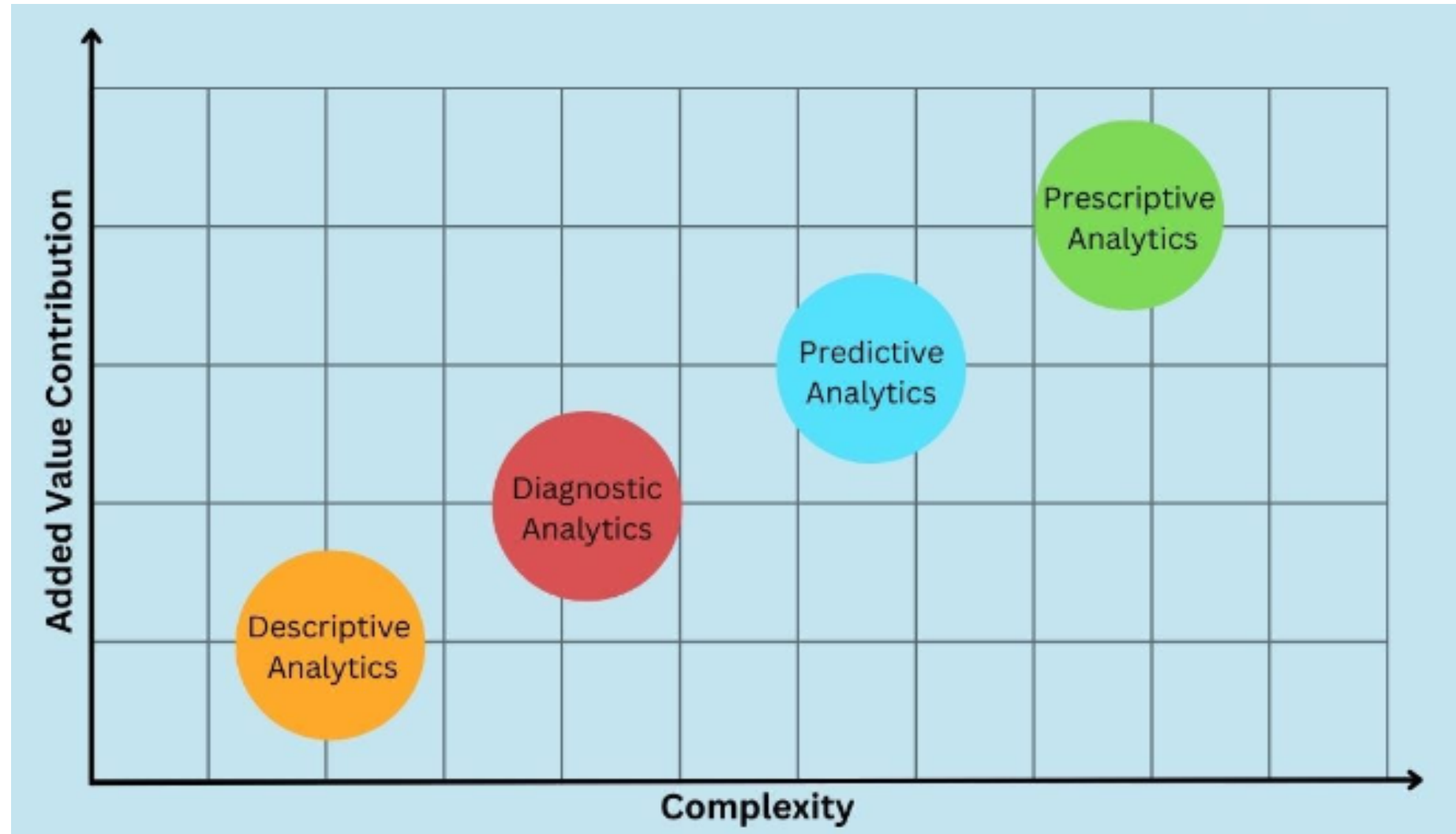
Statistics for Data Analysis

Descriptive Statistics

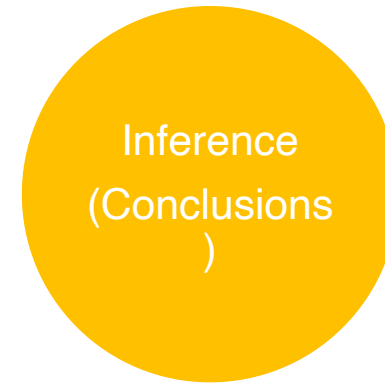
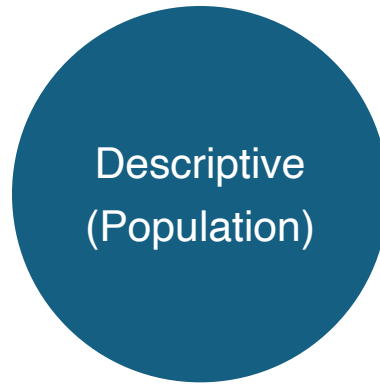
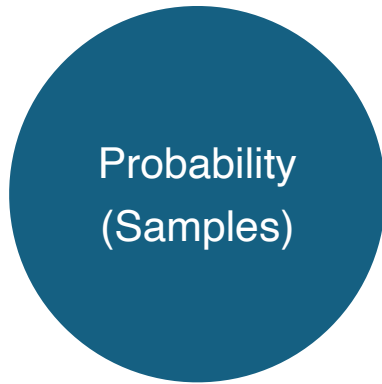
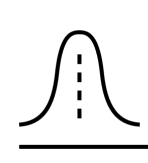
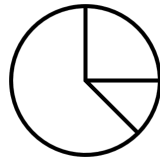
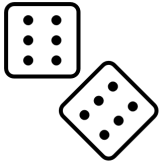
Lecturer: Cristiani Eccher PhD
ceccher@cct.ie



Phases of Data Analytics



Inferential Statistics



Descriptive Statistics

Central Tendency Measures

➤ Mean : $\bar{x}$

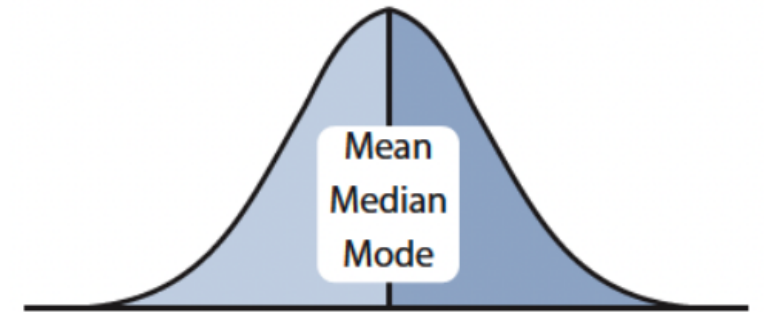
This is the typical average. The calculation is pretty straight forward. We need to sum all the values and divide that result on the total of observations.



Formula

$$\bar{x} = \frac{\sum x}{n}$$

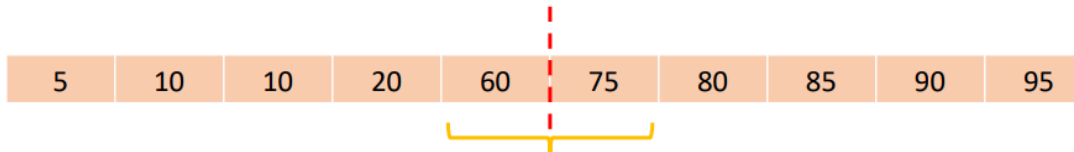
$$\bar{x} = \frac{5 + 10 + 10 + 20 + 60 + 75 + 80 + 85 + 90 + 95}{10} = 53 \text{ years old}$$



➤ Median : Me

This is the value located exactly in the middle of the ordered sequence of values.

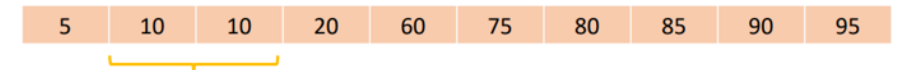
We sort the data from the smallest to the largest and we “cut” the sequence in the middle. That value is the Median.



$$Me = \frac{60 + 75}{2} = 67.5$$

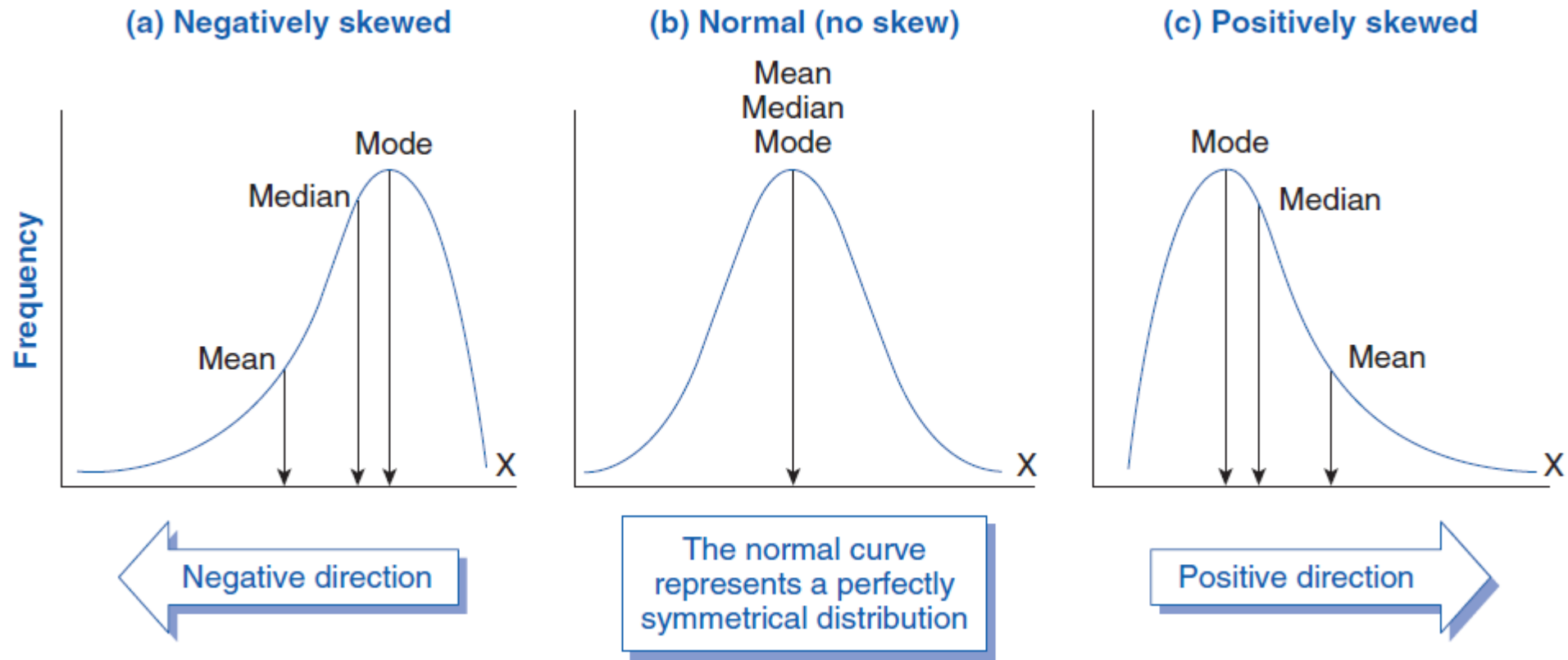
➤ Mode : Mo

This is the number that presents more repetitions within the sample, regardless of how many times it appears.



Mo = 10 years old

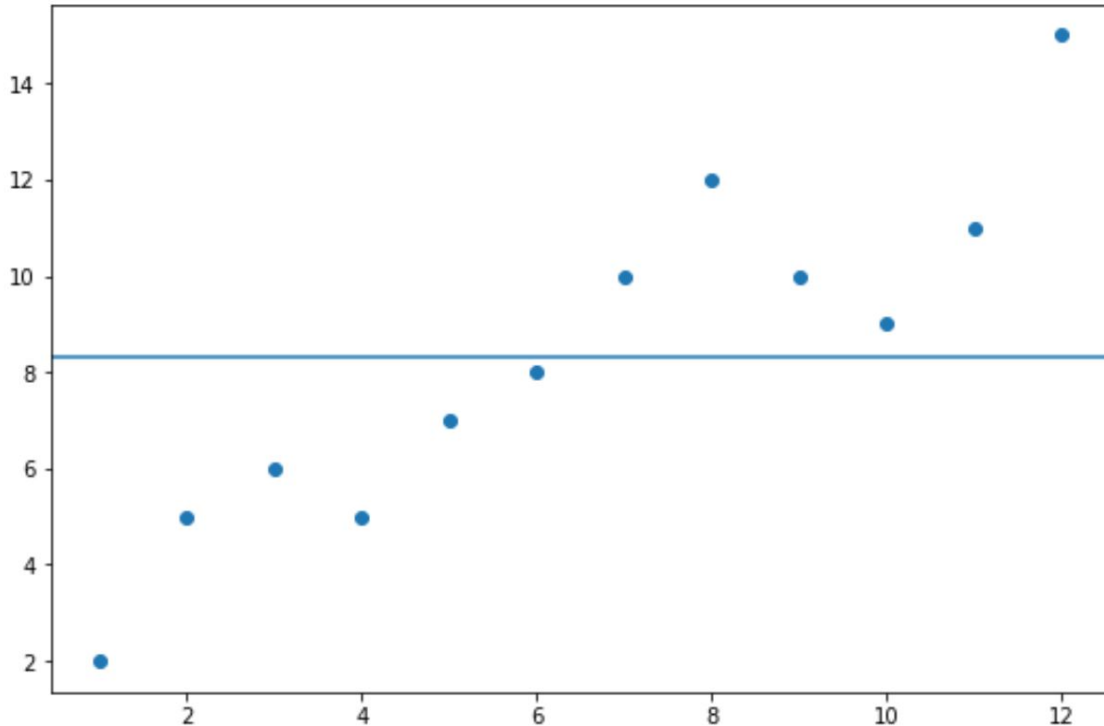
Descriptive Statistics



Descriptive Statistics

Measures of Dispersion

- Variance: The variance is a measure of variability. It is calculated by taking the average of squared deviations from the mean.



$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

x_i = elements in population

μ = population mean

N = population size

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

x_i = elements in sample

$\bar{x}$ = sample mean

n = sample size

Descriptive Statistics

Measures of Dispersion

➤ Variance : S^2

x	$\bar{x}$	$(x - \bar{x})$	$(x - \bar{x})^2$
5	53	-48	2304
10	53	-43	1849
10	53	-43	1849
20	53	-33	1089
60	53	7	49
75	53	22	484
80	53	27	729
85	53	32	1024
90	53	37	1369
95	53	42	1764
			12510



Formula

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

$$S^2 = \frac{12510}{9} = 1390 \text{ years}^2$$

Answer: The variance is 1390 years²

Descriptive Statistics

Measures of Dispersion

➤ Standard Deviation : S

This is the square root of the variance. With the standard deviation we go back to unit of measurement, and we will be able to see the difference of the values respect the mean.

$$S = \sqrt{1390 \text{ years}^2} = 37.28 \text{ years}$$

Answer: The average age of the infected people is 53 years old +- 37,28 years.

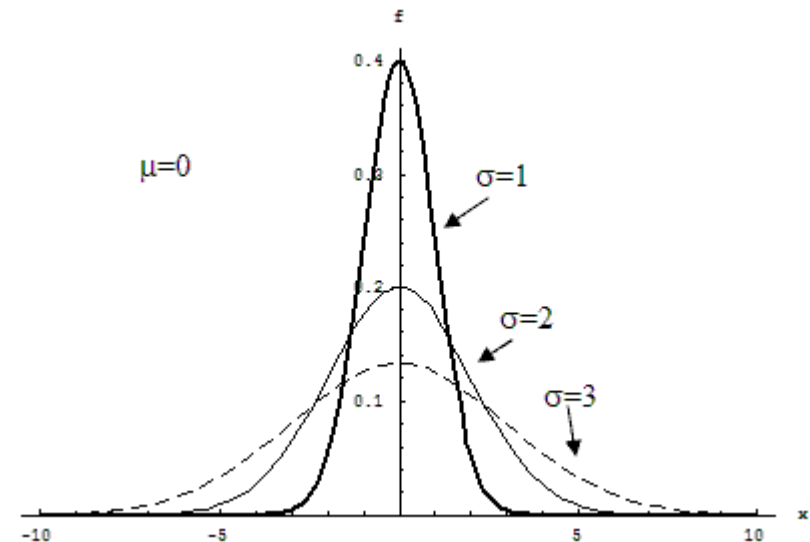
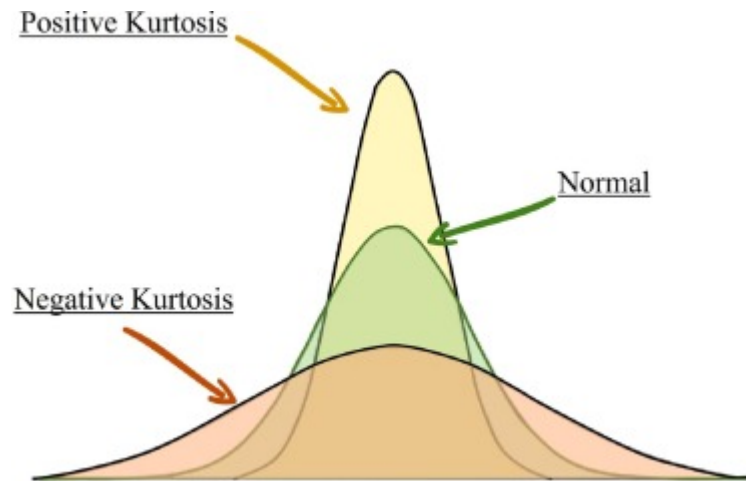
$$VC = S / \bar{X} * 100 =$$

$$VC = (37.28/53) * 100 = 70.33\%$$

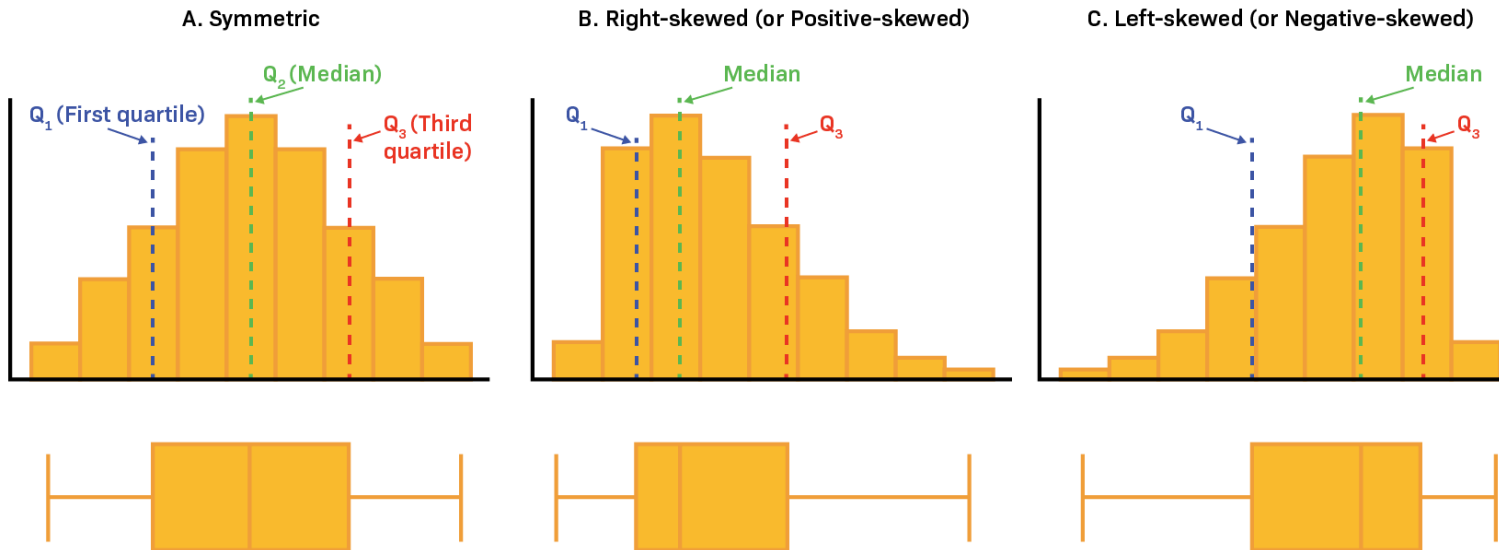
Descriptive Statistics

Measures of Dispersion

- Kurtosis: is a statistical measure that defines how heavily the tails of a distribution differ from the tails of a normal distribution.



Quartiles / Visuals....



Box plot

